

RESEARCH PAPER PRESENTATION

# KnowSelf

Teaching LLM Agents When to Think, Reflect, or Seek Knowledge through  
Situational Self-Awareness

[arxiv.org/abs/2504.03553](https://arxiv.org/abs/2504.03553)



## Current Agent Training Uses "Flood Irrigation"

Existing approaches indiscriminately inject gold trajectories, external feedback, or domain knowledge — regardless of whether the agent needs them at each step.

### Trajectory Imitation

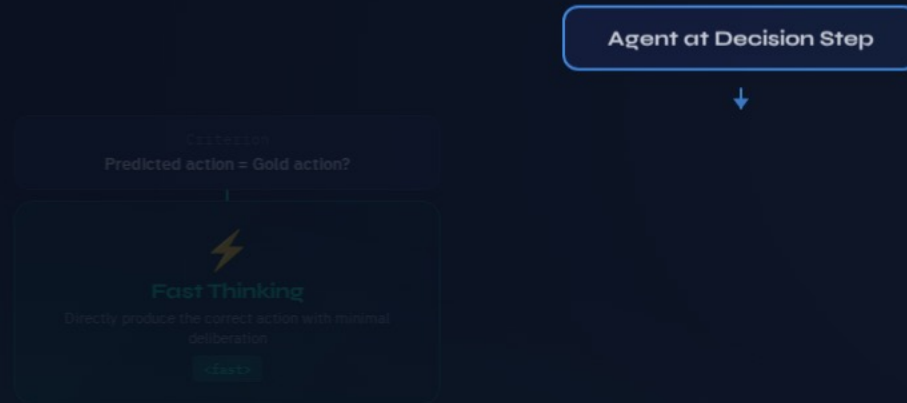
e.g. ReAct

Blindly copies gold action sequences — fragile when inputs deviate from training distribution.

01



## Three Situational Thinking Modes



# KnowSelf Framework: Data to Deployment

STEP 1

Self-Awareness Data Construction

Agent **self-explores** the environment. At each step, heuristic criteria classify the situation and **insert special tokens** (<fast>, <reflection>, <knowledge>) into trajectories.



# KnowSelf Correctly Acts Where Baselines Fail

Task: "Put a hot mug in cabinet"

**OpenAI-O1** Incorrectly tries to take an egg from the microwave after opening it — fails to heat the mug

**KnowSelf** Uses `<reflection>` token to realize it should heat the mug using the open microwave — succeeds



# KnowSelf Achieves 84.33% on ALFWorld (Llama-8B) with Only 15% Knowledge

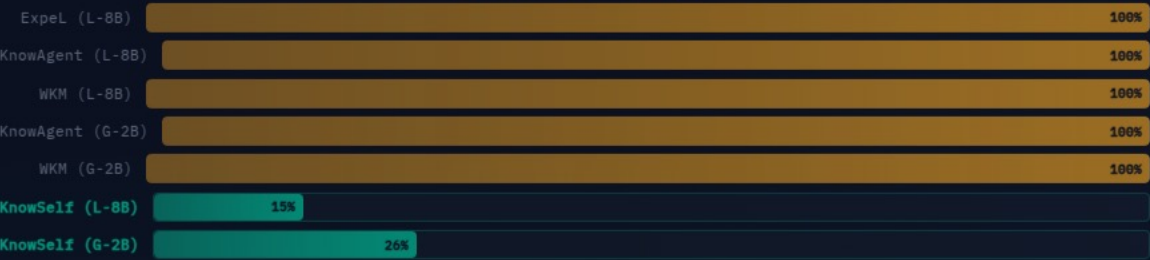
| Backbone | Method    | Know%  | Put    | Clean | Heat  | Cool  | Examine | Two   | All   |
|----------|-----------|--------|--------|-------|-------|-------|---------|-------|-------|
| GPT-4o   |           |        |        |       |       |       |         |       |       |
|          | ReAct     | 0%     | 83.33  | 74.19 | 69.57 | 85.71 | 77.78   | 64.71 | 76.12 |
|          | Reflexion | 0%     | 100.00 | 87.10 | 73.91 | 90.48 | 83.33   | 70.59 | 85.07 |
|          | ExpeL     | 100%   | 95.83  | 83.87 | 69.57 | 80.95 | 88.89   | 52.94 | 79.85 |
| Llama-8B |           |        |        |       |       |       |         |       |       |
|          | ReAct     | 0%     | 33.33  | 3.23  | 0.00  | 57.14 | 66.67   | 23.53 | 27.61 |
|          | Reflexion | 0%     | 62.96  | 22.73 | 5.88  | 64.29 | 86.36   | 50.00 | 51.49 |
|          | ExpeL     | 100%   | 83.33  | 32.26 | 30.43 | 23.81 | 55.56   | 17.65 | 41.04 |
|          | ETO       | 0%     | 91.67  | 70.59 | 82.61 | 61.90 | 88.89   | 64.71 | 78.36 |
|          | KnowAgent | 100%   | 87.50  | 93.55 | 65.22 | 66.67 | 61.11   | 64.71 | 75.37 |
|          | WKM       | 100%   | 95.83  | 87.10 | 86.96 | 61.90 | 66.67   | 52.94 | 77.61 |
|          | KnowSelf  | 15.01% | 91.67  | 87.10 | 91.30 | 85.71 | 77.78   | 64.71 | 84.33 |
| Gemma-2B |           |        |        |       |       |       |         |       |       |
|          | ReAct     | 0%     | 0.00   | 9.68  | 0.00  | 4.76  | 44.44   | 0.00  | 8.96  |
|          | ETO       | 0%     | 91.67  | 83.87 | 78.26 | 52.38 | 77.78   | 29.41 | 71.64 |
|          | KnowAgent | 100%   | 91.67  | 90.32 | 69.57 | 71.43 | 66.67   | 41.18 | 73.88 |
|          | WKM       | 100%   | 91.67  | 87.10 | 78.26 | 71.43 | 61.11   | 52.94 | 76.12 |
|          | KnowSelf  | 26.41% | 87.50  | 93.55 | 73.91 | 76.19 | 83.33   | 52.94 | 79.85 |

Llama-8B KnowSelf 84.33% > GPT-4o ExpeL 79.85%

Gemma-2B KnowSelf 79.85% = GPT-4o ExpeL

# Minimal Knowledge, Maximum Impact

Knowledge Usage (%)



Overall Accuracy (%)



## Key Takeaways

**01** Indiscriminate knowledge injection is wasteful and suboptimal — flooding agents with knowledge at every step degrades performance.

**02** Agents can learn situational self-awareness via special tokens with lightweight heuristic labeling — no expensive human annotation needed.

